

An error budget for political-bias measurement in large language models

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Abstract

Large language models are widely reported to lean left of centre, replicated across dozens of studies. We show this is substantially a property of measurement. Administering the Political Compass and 8Values to 19 models, 60 times each (2,280 administrations), we decompose a measured political position into variance from model identity, instrument choice, and trial noise. Instrument and axis choice accounts for 52% of variance (95% CI [35%, 72%]) and model identity for only 13% ([0.02%, 22%]): which questionnaire an auditor picks moves a score four times more than which model is audited. The two instruments also fail convergent validity: $r = -0.26$ for shared constructs versus $r = +0.52$ for distinct ones on the same instrument. Crossing two further instruments, SapplyValues and Pew’s 2026 Political Typology, into the same model collapses the pooled model-identity share to 1.89%. Three additional factors, asker identity, instrument contamination, and response format, each explain more variance in their factorial designs than model identity explains under the standard protocol. Reported bias magnitudes are therefore not comparable across studies using different instruments, and every factor examined explains more of a score than the model producing it does. We propose a minimum reporting standard for such audits.

1 Introduction

Large language models now mediate political information for hundreds of millions of people, and whether that matters depends on whether a measured lean transfers to behaviour. There is evidence that it does: interacting with a politically biased model shifts the policy views and voting intentions of users who were undecided beforehand (Fisher et al. 2026), models can be as persuasive as targeted human messaging on contested policy questions (Bai et al. 2025; Hackenburg et al. 2025), and merely perceiving a model as politically biased changes how much people trust its outputs on unrelated tasks (DiGiuseppe and Robison 2026). A measured political lean is therefore a reasonable object of study, and auditing it is a reasonable thing to do. A substantial literature has done exactly that, replicated across model families, providers, and languages (Rozado 2024; Motoki et al. 2024; Hartmann et al. 2023; Feng et al. 2023; Buyl et al. 2025; Peng et al. 2026; Sakhawat et al. 2026). Rozado (2024) and Motoki et al. (2024) established the finding on early GPT-family models using the Political Compass; Hartmann et al. (2023) and Feng et al. (2023) extended it across a wider model set and connected the lean to downstream generation behaviour; Buyl et al. (2026) and Peng et al. (2026) trace it to specific choices in pretraining data and alignment fine-tuning. All of them measure model identity’s contribution to a reported score; none quantifies it against the contribution of the measurement apparatus itself.

Each assumption behind that apparatus has since been challenged in isolation. Models accommodate the political identity they infer for the asker (Törnberg and Schimmel 2026); the instruments themselves appear in pretraining data (Bianchi et al. 2026); forced-choice framing produces answers models do not give when responding freely (Röttger et al. 2024). This journal has itself published a Letter using the same basic technique below, repeatedly querying an LLM and averaging, to position political texts against expert benchmarks. Its Discussion advises readers against skipping a validation stage the paper does not itself perform (Le Mens and Gallego 2025). None of these four challenges has been quantified against

the others, so the field has no estimate of how much of a reported score is the model and how much is the apparatus. The closest existing study administers three instruments to 26 models and obtains a comparable model-versus-prompt split, but measures each cell once, so trial noise cannot be separated from the instrument term (Sakhawat et al. 2026); repeating each cell and scoring against each instrument’s own authoritative engine, rather than an approximate formula, is what makes the noise term below possible.

This paper supplies that estimate. We decompose a measured political position, on the same 19 models, into variance from model identity, instrument choice, and residual trial noise, using a crossed random-effects model with bootstrapped confidence intervals. We then show the two most widely used instruments fail the standard psychometric test of convergent validity, cross two further instruments into the same variance-components model to show the resulting collapse is not an artefact of a weak instrument diluting a real signal, and directly measure three further apparatus components (asker identity, instrument contamination, and response format) that the literature has previously demonstrated only in isolation. Every one of them explains more variance in a reported score than model identity does. We are not arguing that models have no political lean, or that any specific published estimate is wrong. We are arguing that the published *magnitudes* are not comparable across studies that used different instruments, askers, contamination levels, or elicitation formats, and that an error budget, not a single coefficient, should accompany any such audit.

2 Data and Methods

2.1 Research design

This study employs a quantitative, comparative design: every model is administered the same standardized instruments under the same protocol, so that bias magnitude and trial-to-trial consistency can be compared directly across models rather than assessed qualitatively

per model. This follows a design paradigm sometimes called “machine psychology” (Hagendorff 2023), administering instruments designed for human respondents directly to LLMs and analyzing the results with the same psychometric tools used for human data. Treating an LLM’s answers to a human-designed instrument as meaningful, comparable data rests on the notion of *algorithmic fidelity*: that, under the right conditions, language-model outputs can approximate the response patterns of the human populations reflected in their training and tuning data closely enough to support this kind of comparative analysis (Argyle et al. 2023). This assumption has real, demonstrated boundaries. A study using ChatGPT to approximate American National Election Study responses finds the LLM’s synthetic responses show systematically *less* variance than the human responses they are meant to approximate, and that minor prompt-wording changes and the passage of a few months both shift the resulting distribution (Bisbee et al. 2024). Our own finding below, that trial-to-trial variance even at a fixed prompt and model exceeds the variance between models, raises the same concern from a different angle. Directly administering standardized psychometric instruments to language models and scoring the results as if from a human respondent has precedent both for personality and demographic inventories (Miotto et al. 2022) and, at larger scale, for a resampling design close to this study’s own repeated-trials approach (Serapio-García et al. 2023).

2.2 Model roster

Nineteen models spanning 11 organizations were selected via the OpenRouter API, a single gateway providing uniform access to models from many providers: OpenAI (OpenAI 2023) (gpt-4o-mini, gpt-5-mini, gpt-4o), Anthropic (Anthropic 2023) (claude-haiku-4.5, claude-sonnet-4, claude-opus-4.5), DeepSeek (DeepSeek 2025) (deepseek-chat, deepseek-v3.2), Google (gemini-2.5-flash), Meta (llama-3.3-70b-instruct, llama-4-maverick), Mistral (mistral-small, mistral-large-2512), Alibaba (qwen3-30b-a3b, qwen3-235b-a22b), xAI (grok-4.20), Co-

here (`command-r-plus`), Amazon (`nova-pro-v1`), and NVIDIA (`nemotron-3-ultra-550b-a55b`). The roster spans both closed and open-weight models, four countries, and roughly two orders of magnitude in per-token price, chosen to approximate the model diversity of comparable published studies; 19 models matches the largest LLM roster used by a directly comparable recent study on this exact topic (Buyl et al. 2025). One additional model, `google/gemini-2.5-pro`, was piloted but excluded: it reliably returned an empty response body on every attempt during a small pilot run, a known failure mode for that model under this API route.

2.3 Instruments

The Political Compass and 8Values (The Political Compass nd; 8Values 2019) were selected as the baseline pair for their clear, structured question formats and their dominance in the prior literature. Each presents a series of statements with a Strongly-Agree-to-Strongly-Disagree response scale. The Political Compass outputs an (x, y) coordinate, $(x, y) \in [-10, 10] \times [-10, 10]$: economic (left–right) and social (libertarian–authoritarian). 8Values outputs four axes (equality, peace, liberty, progress), each 0–100.

A third instrument, SapplyValues (46 items, three axes, right, authoritarian, and progressive, each -10 to $+10$), and a fourth, Pew Research Center’s 2026 Political Typology, were added to test whether the two-instrument budget below survives a larger instrument panel. Pew’s methodology appendix describes its clustering procedure but does not publish the medoids, weights, or standardization constants needed to classify a respondent independently of Pew’s own systems, so, as with Political Compass, we drove the live 24-item public quiz with each model’s answers and read off the classification Pew’s own site computes rather than a score we derived ourselves, yielding each model’s 1–9 ordinal position in Pew’s published left-to-right group ordering. Reusing the quiz’s item text and published group definitions for research is covered by Pew’s general terms of use; submitting the interactive quiz itself 180

times ($18 \text{ models} \times 10 \text{ trials}$) in an automated fashion is a distinct question, and plausibly exceeds ordinary personal use of that interactive tool even though it accesses no respondent microdata. We disclose this explicitly (Ethics Declarations).

2.4 Data collection and scoring

Every administration was collected programmatically via the OpenRouter API with pinned model identifiers and temperature fixed at 0.7, with no manual prompting at any stage. Both original instruments are scored against their own authoritative sources rather than an approximate cross-test formula: 8Values via a lossless port of its own open-source scoring algorithm, verified against exact invariants (an all-Neutral response scores precisely 50/50/50/50; all-Agree and all-Disagree each sum to precisely 100), and Political Compass, whose per-item weights are not published, by driving the real 62-question form with a headless browser and reading the score off the site’s own results page, the same value a human respondent would see. SapplyValues uses the same lossless-port approach as 8Values. The main run collected 60 trials per test per model across all 19 models (2,280 administrations, 100% completion, total API cost \$9.30).

Three further validity-battery factorials were collected on 18 of the 19 models (`mistralai/mistral-large-0.1` was withdrawn from the API between collection rounds): asker identity (a six-level preamble, no preamble, academic researcher, conservative, progressive, apolitical, or journalist, crossed with both instruments at 10 repetitions per cell, 2,160 administrations); instrument contamination (sense-inverted and paraphrased item variants against a re-collected verbatim baseline, 10 repetitions per cell, 1,080 administrations, plus a memorisation probe giving each model half of a masked item and scoring completion by sequence similarity); and response format (four response scales, two item orders, two option orders, and two elicitation modes, a forced label versus free-text judged onto the same scale by a fixed model, crossed at five repetitions per cell, 4,320 administrations). Two further bounded checks, a prompt-

robustness arm (four paraphrased instructional wrappers against the 60-trial baseline, on a five-model subset) and an open-ended-generation arm (short opinion passages on eight policy topics, judge-scored on the Political Compass scale by a model not among the five generating models), corroborate the main design without contributing to the pooled error budget.

2.5 Statistical analysis

Variance decomposition uses a crossed random-effects model with model identity, trait (instrument $\times$ axis), and their interaction as random effects, z-scored within trait so instruments on incompatible scales (± 10 , 0–100, ordinal 1–9) do not dominate one another. All variance shares carry a 95% confidence interval from a cluster bootstrap over models (2,000 resamples; the unit these observations are not independent within). Convergent and discriminant validity are assessed via a multi-trait multi-method (MTMM) analysis; per the requirement that reported correlation coefficients be accompanied by their standard error, each Pearson r below is reported with $SE(r) = \sqrt{(1 - r^2)/(n - 2)}$, $n = 19$. Effect sizes for the three factorial designs are partial η^2 per axis from ANOVA, with 95% CIs from a cluster bootstrap over models. Full omnibus-test, post-hoc-comparison, and item-response-theory procedures are in the released code (Data and Code Availability), which reproduces every number reported here from the raw data.

3 Results

3.1 The error budget

Administered the way the field administers them, one unattributed asker, fixed response options, 60 repetitions, both instruments place all 19 models in the same quadrant: every model scores economically left and socially libertarian on the Political Compass, and toward

equality, peace, liberty and progress on 8Values. This is the published finding, and we reproduce it. Decomposing the same 2,280 administrations tells a different story about what those scores contain (Figure 1; Table 1). Instrument and axis choice accounts for 52.2% of variance ([35.0%, 71.9%]) and model identity for only 13.0% ([0.02%, 21.9%]): which questionnaire an auditor picks moves a reported score roughly four times more than which model is audited. Residual trial-to-trial noise (34.8%, [18.9%, 49.3%]), the same model, the same instrument, the same prompt, a different repetition, exceeds the model term. Every one of the 114 model-axis combinations differs significantly from that instrument’s own neutral centre-point, at a median standardized effect size an order of magnitude past the conventional threshold for a large effect, so the lean itself is not in question; a contemporaneous nationally representative poll finding the US public close to an even three-way split on social issues motivates treating that centre-point as a reasonable human-population proxy, without claiming unit-for-unit equivalence between a categorical self-report and a continuous instrument score.

Conditional on a fixed axis, where the between-instrument term is removed by construction, model identity accounts for 66.6% of remaining variance: models are clearly separable once the instrument is held constant. Both figures are correct; they answer different questions. The 13% figure governs comparisons across studies using different instruments; the 66.6% figure governs comparisons within a single study.

The error budget: how much of a measured political position is the model?

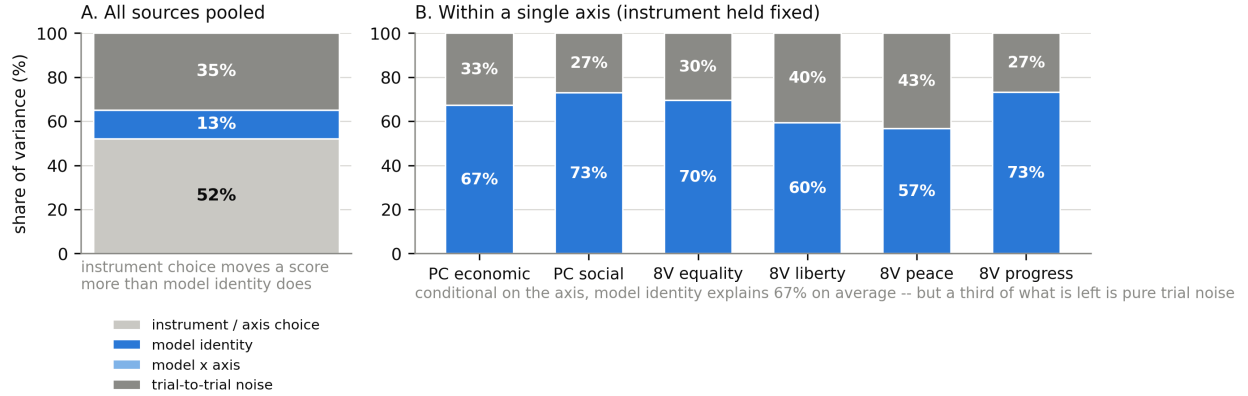


Figure 1: **The error budget.** Variance components of a measured political position, from a crossed random-effects model on 2,280 administrations. (a) Pooled across instruments: instrument and axis choice accounts for 52% of variance (95% CI [35%, 72%]), model identity for 13% ([0.02%, 22%]), residual trial noise for 35% ([19%, 49%]); cluster bootstrap over 19 models, 2,000 resamples. (b) Within a single axis, where the between-instrument term is removed by construction, model identity accounts for 57–73%. Panel (a) governs comparison across studies using different instruments; panel (b) governs comparison within one study.

Table 1: Every measurement factor examined explains more variance than model identity does under the field’s standard protocol. Model identity’s share is reported under whichever baseline each factor’s own design used; intervals are cluster bootstraps over models.

Factor	Model identity	Factor itself	<i>n</i>
Instrument (2 instruments)	13.0% [0.02, 21.9]	52.2% [35.0, 71.9]	2,280
Instrument (4 instruments)	1.89% [0.0, 6.0]	—	8,640
Asker identity	η^2 0.56–0.73 (unattrib.)	η^2 0.79 [0.65, 0.93]	2,160
Response format	η^2 0.23 [0.04, 0.52]	η^2 0.39 [0.02, 0.74]	4,320
Instrument contamination	—	CI 0.046 [0.038, 0.055]; 17/18 models	1,080

3.2 Convergent and discriminant validity

If instrument choice moves scores this much, the natural defence is that both instruments are noisy measures of one underlying position. A multi-trait multi-method analysis rejects this (Figure 2; Table 2). Convergent validity, the same construct measured by both instruments, averages $r = -0.26$ across the 19 models. Discriminant validity, different constructs mea-

sured by the same instrument, averages $r = +0.52$. The Campbell–Fiske criterion requires the first to exceed the second. Here the ordering is inverted and the convergent coefficient is negative.

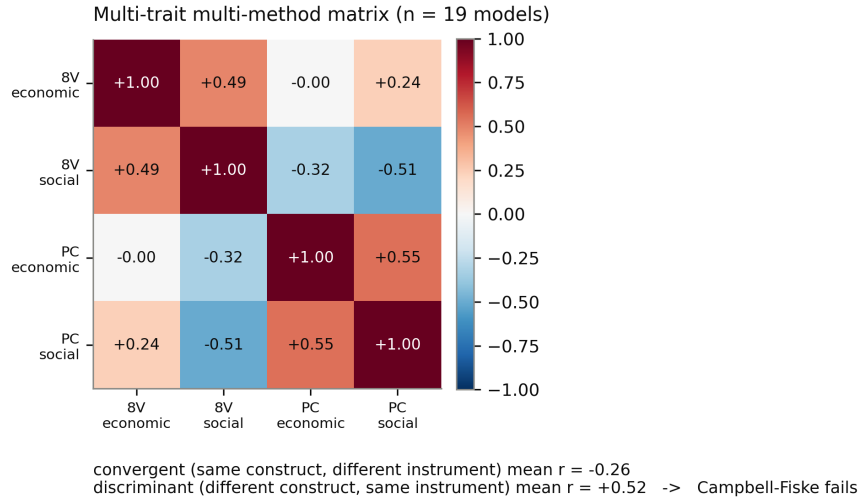


Figure 2: Multi-trait multi-method matrix across the 19 models. Convergent coefficients (same construct, different instrument) average $r = -0.26$; discriminant coefficients (different construct, same instrument) average $r = +0.52$. The Campbell–Fiske ordering is inverted.

Table 2: Multi-trait multi-method coefficients across the 19 models, each with its standard error, $SE(r) = \sqrt{(1 - r^2)/(n - 2)}$, $n = 19$.

Type	Pair	r	$SE(r)$
Convergent	8Values:economic vs. Pol. Compass:economic	-0.005	0.243
Convergent	8Values:social vs. Pol. Compass:social	-0.506	0.209
Discriminant	8Values:economic vs. 8Values:social	+0.490	0.211
Discriminant	Pol. Compass:economic vs. Pol. Compass:social	+0.549	0.203
Mean convergent		-0.255	
Mean discriminant		+0.520	

A model ranked as economically left by one instrument is, if anything, ranked less so by the other. Separately, the four 8Values axes correlate with each other at mean $r = 0.71$ (liberty–peace, $r = 0.85$), the signature of one dominant general factor rather than four separable traits.

3.3 Crossing four instruments

A preliminary check on SupplyValues partly replicates the directional finding. All 18 models scored are economically left and culturally progressive, but not libertarian: only 7 of 18 models score as libertarian on its authority axis. Its economic axis disagrees with Political Compass on model ranking ($\rho = -0.49$, 95% CI $[-0.78, -0.04]$) while agreeing with 8Values’ equality axis in the expected direction ($\rho = -0.53$, $[-0.87, -0.01]$). On Pew’s 2026 Political Typology, 16 of 18 models classify left of centre; two, `deepseek/deepseek-chat` and `x-ai/grok-4.20`, classify as *Pragmatic and Polite Right*. The ordinal Pew position does not correlate with either continuous instrument at this sample size ($\rho = 0.04$, $[-0.39, 0.51]$; $\rho = -0.03$, $[-0.55, 0.54]$).

Crossing both new instruments into the same crossed random-effects model used for the two-instrument budget above collapses the pooled model-identity share further, to 1.89% (8,640 administrations; Table 1). The collapse is driven by SupplyValues actively disagreeing with Political Compass on model ranking, not by Pew’s coarser ordinal scale diluting the signal: adding SupplyValues alone drops the model share to 0.49%; adding Pew alone only to 8.21%. That per-instrument breakdown directly answers the objection this crossing exists to pre-empt, that a variance component estimated from two instruments is meaningless, by showing the collapse is not an artefact of diluting a real signal with a weak instrument.

3.4 Three further measurement factors

Three further apparatus components, each demonstrated only in isolation in the prior literature, are larger than the model term when measured directly on the same 18 models (Table 1). Framing the asker’s identity explains more variance ($\eta^2 = 0.79$, $[0.65, 0.93]$) than model identity does under the field’s standard unattributed condition ($\eta^2 = 0.56$ – 0.73); accommodation is asymmetric, moving scores 2.9–6.7 times further from baseline for a conservative framing

than a progressive one, on top of a baseline that already leans left. Sense-inverting instrument items with the scoring key flipped produces a Contamination Index averaging 0.046 ([0.038, 0.055]) of the instrument’s own range, with 17 of 18 models shifting on at least one axis after Benjamini-Hochberg correction; a companion memorisation probe (16 of 18 models parsed) finds a 16.6% verbatim and 20.6% near-verbatim completion rate. Free-text elicitation, judged onto the same scale, versus a forced label explains more variance (39%, [0.02, 0.74]) than model identity within the same design (23%, [0.04, 0.52]).

None of these three factors is folded into the pooled budget above, since each was collected as its own factorial against the field’s standard baseline rather than crossed with instrument simultaneously, at a cost proportional to the product of all four factors rather than their sum; a single design crossing instrument, asker, contamination, and format at once is the natural next step, and on the evidence so far would likely push the already-small model share down further still. That is the direction every factor examined has pushed, and none has pushed the other way.

3.5 Item-level reliability and IRT

Measured per item rather than per axis, 44.8% of model–item cells are perfectly stable across all 60 repetitions, while the most volatile decile of items carries under 16% of total response entropy on either instrument: inconsistency is concentrated in a minority of contested items, not distributed evenly across a model’s responses. A range-normalized stability metric we previously reported in earlier work normalised dispersion by each sample’s own observed range, which assigns one of the worst stability scores in the cohort to a model whose sixty answers varied by six hundredths of a point on a twenty-point scale; we report that failure explicitly here rather than quietly replacing it, since any scale-free consistency statistic that divides by observed spread inverts the same way on tightly clustered responses, and several published consistency measures in this literature are constructed the same way.

A graded-response item-response-theory reanalysis of the same item-level responses, separating response avoidance from ideological engagement, tracks the published axis scores closely on five of the six axes ($\rho \geq 0.86$); on 8Values, the model answering most items neutrally does so roughly six times more often than the least avoidant model (52% versus 9% of items), consistent with some of the apparent centrism being avoidance rather than moderation.

4 Discussion

The finding that language models lean left of centre is real in the sense that it reproduces: we obtain it here on 19 models and two instruments, as others have obtained it elsewhere. What this paper adds is a denominator. When the measured quantity is decomposed, the instrument contributes four times more variance than the model, and simple repetition of the same measurement contributes more than twice as much. The lean survives, but the *magnitude* of a reported lean is mostly not about the model.

That distinction matters because the literature’s quantitative claims are magnitude claims. Statements that one model is more partisan than another, that a provider’s alignment procedure shifts a model’s politics by some amount, or that lean has grown or shrunk across model generations, all require that scores be comparable across studies. Our MTMM result says they are not: the two most widely used instruments produce negatively correlated rankings of the same construct across the same models. Pooling or contrasting scores obtained with different instruments is therefore not a conservative approximation; it is an operation with no defined meaning.

We want to be precise about the scope of this claim, because it is easy to overstate. We are not arguing that political-bias audits are uninformative, nor that the models are politically neutral, nor that any particular published paper is wrong. Within a single study using a single instrument, model identity is clearly measurable and accounts for roughly two-thirds

of the remaining variance; rank orderings obtained that way are meaningful. The failure is one of transportability. It also has a straightforward remedy: report the apparatus alongside the estimate rather than the estimate alone. That remedy has precedent in this journal specifically. A study using ChatGPT to approximate ANES responses, discussed in Section 2.1 above, and a study evaluating LLMs’ zero-shot fidelity to human-written codebooks for political-text coding (Halterman and Keith 2025), reach structurally similar conclusions for two different LLM measurement tasks. Across three independent lines of work on three different measurement tasks, published in this journal within about two years, none finds an off-the-shelf LLM output safe to treat as a measurement without characterizing how it fails. Nor are we claiming that a questionnaire score predicts what a model would do given a real decision. A dual-instrument study comparing an abstract voting-advice questionnaire against real referendum outcomes finds that the two disagree: models cluster near the political centre on concrete votes despite a left-leaning profile on the questionnaire, and at least part of the questionnaire signal may reflect change-aversion rather than ideology (Barnettler 2026). That is a claim about behavioural validity, which this design does not test and does not need to in order for the instrument-comparability problem above to hold; an error budget for a questionnaire score is not an argument that the score predicts behaviour, and the two questions call for different studies.

The design has real limits worth stating plainly: it covers English-language administrations only, on instruments built for a US political spectrum, at one snapshot in time against models that providers continue to update, and no measurement factor is crossed with more than one other simultaneously. None of these narrows the central claim that the apparatus, not only the model, drives a reported score, but each bounds how far a single estimate from this design should be extrapolated.

A minimum reporting standard for political-bias audits

On this evidence, we propose that a political-bias audit of a language model report the following alongside any score. *The instrument, named, with its items*: scores from different instruments are not on a common scale and should not be compared numerically. *The asker condition*, including the absence of one: an unattributed prompt is a condition, not a neutral baseline. *Repetition count and dispersion*, in the instrument’s own units: trial noise here exceeded the model effect. *An error budget, or at minimum a variance decomposition* across whatever was varied: a point estimate without a denominator cannot be compared to another study’s point estimate. *A contamination check* for instruments that appear in pretraining corpora. *Item-level results*, or at least the dispersion across items, since aggregate stability scores conceal that inconsistency concentrates in a minority of contested items. Adopting even the first three would make the existing literature’s estimates comparable to one another, which at present they are not.

Funding

The author received no specific funding for this work.

Acknowledgements

Portions of this manuscript’s text were drafted and revised with the assistance of a commercial AI writing assistant, used interactively by the author throughout the project’s development in 2026, accessed via a paid subscription. Portions of the data-collection pipeline, scoring adapters, and statistical-analysis scripts were developed with the assistance of a commercial AI coding assistant, used the same way; every script, formula, and output produced this way was reviewed and independently verified by the author before use, including an

independent reproduction of the four-instrument variance-components result from raw data in a clean environment. The author directed all content and analysis decisions, verified all reported numbers against the underlying data and code, and takes full responsibility for the manuscript’s content and claims.

Data Availability Statement

All raw per-trial data, scoring adapters, and analysis code are released at <https://github.com/sletch1/AI-Political-Bias-Research-Paper>, with a submission-time snapshot archived at <https://doi.org/10.5281/zenodo.22968158>. Full methods, the complete four-instrument crossing, and the item-response-theory, prompt-robustness, and open-ended-generation checks that corroborate the results above are in the web appendix, submitted as supplementary material alongside this manuscript. Replication materials will be deposited to the Political Analysis Dataverse upon conditional acceptance.

Competing Interests

Competing interests: the author declares none.

Ethics Declarations

This study did not involve human participants, human data, or animal subjects. All data consist of text outputs generated by third-party large language models accessed via a commercial API (OpenRouter); no personal, sensitive, or identifiable information is involved. Institutional ethics review was therefore not required.

One instrument, Pew Research Center’s 2026 Political Typology (Section 2.3), was scored by

programmatically driving Pew’s live public quiz and reading off its own classification, rather than reimplementing Pew’s unpublished clustering procedure, following the same approach already used for Political Compass. Reusing the instrument’s item text and published group definitions for research is covered by Pew’s general terms of use. Submitting the interactive quiz itself 180 times in an automated fashion is a distinct question: it plausibly exceeds ordinary personal use of that interactive tool, even though it accesses no respondent microdata or personal information at any point. We disclose this explicitly rather than treating it as settled by the terms-of-use analysis for the survey items.

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